STT 465: Bayesian Data Analysis

Homework 3

Normal Models, Prior Choice, and Unknown Variance

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Total: 5 points **Due: 10/30 5PM**

Instructions.

Show all essential derivations and justify your conclusions. A numerical answer without the corresponding Bayesian argument may receive limited credit.

You may use `R` for numerical calculations, simulation, and plotting. For computational questions, include the relevant code and briefly interpret the results.

Question 1: Normal Model with Known Variance [1 point]

Suppose

$$X_1, \ldots, X_n \mid \mu \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

where σ^2 is known, and suppose

$$\mu \sim N(\mu_0, \tau_0^2).$$

Let

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

(a) Starting from the likelihood and prior kernels, derive

$$\mu \mid \boldsymbol{x} \sim N(\mu_n, \tau_n^2),$$

where

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1},$$

and

$$\mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}\right).$$

Your derivation should identify the posterior precision.

[0.20 pt]

(b) Show that

$$\mu_n = w\bar{x} + (1-w)\mu_0$$

for some $0 < w < 1$, and derive w.

Explain how the relative prior and data precisions determine w.

[0.15 pt]

(c) Derive the posterior predictive distribution of a future observation X_f.

[0.20 pt]

(d) Suppose

$$n = 16, \qquad \bar{x} = 5.4, \qquad \sigma^2 = 4,$$

and

$$\mu \sim N(4, 1).$$

Find the posterior mean, posterior variance, and the posterior predictive distribution.

[0.20 pt]

(e) Repeat part (d) using the diffuse prior

$$\mu \sim N(0, 1000^2).$$

Compare the two posterior means and explain why the difference between them is relatively small.

[0.25 pt]

Question 2: Prior Choice and Reparameterization [1 point]

This question examines how prior information depends on the chosen parameterization.

(a) Suppose an expert believes that a normal mean μ is approximately centered at 30 and that

$$\mathrm{P}(10 \leq \mu \leq 50) \approx 0.95.$$

Using the approximation that about 95% of a normal distribution lies within two standard deviations of its mean, choose μ_0 and τ_0 for the prior

$$\mu \sim N(\mu_0, \tau_0^2).$$

[0.15 pt]

(b) Consider instead the reference prior

$$\pi(\mu) \propto 1, \qquad -\infty < \mu < \infty.$$

Explain why this is an improper prior.

For the model

$$X_1, \ldots, X_n \mid \mu \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

with known σ^2, derive the posterior distribution under this prior and show that it is proper for $n \geq 1$.

[0.25 pt]

(c) Suppose

$$\sigma^2 \sim \text{Unif}(0, 10),$$

and define the precision

$$\widetilde{\sigma} = \frac{1}{\sigma^2}.$$

Using the change-of-variables formula, derive the induced prior density for $\widetilde{\sigma}$, including its support.

[0.20 pt]

(d) Now suppose instead that the reference prior is chosen directly on the precision scale:

$$\pi(\widetilde{\sigma}) \propto 1, \qquad \widetilde{\sigma} > 0.$$

Show that this prior is not equivalent to placing a flat prior on σ^2.

Explain in one or two sentences why the phrase "non-informative prior" can depend on the parameterization.

[0.20 pt]

(e) Jeffreys' prior for the normal precision is

$$\pi_J(\widetilde{\sigma}) \propto \widetilde{\sigma}^{-1}.$$

Transform this prior from $\widetilde{\sigma} = 1/\sigma^2$ to a prior on σ^2.

Comment briefly on why this illustrates the transformation-invariance property of Jeffreys' prior.

[0.20 pt]

Question 3: Normal Model with Known Mean [1 point]

Suppose

$$X_1, \ldots, X_n \mid \sigma^2 \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

where μ is known. Define the precision

$$\widetilde{\sigma} = \frac{1}{\sigma^2}$$

and

$$SS_\mu = \sum_{i=1}^{n} (x_i - \mu)^2.$$

Assume the Gamma distribution uses the **rate** parameterization.

(a) Show that the likelihood for $\widetilde{\sigma}$ has kernel

$$f(\boldsymbol{x} \mid \widetilde{\sigma}) \propto \widetilde{\sigma}^{n/2} \exp\left(-\frac{\widetilde{\sigma}}{2} SS_\mu\right).$$

[0.15 pt]

(b) Suppose

$$\widetilde{\sigma} \sim \text{Gamma}(a_0, b_0).$$

Derive the posterior distribution

$$\widetilde{\sigma} \mid \boldsymbol{x} \sim \text{Gamma}(a_n, b_n)$$

and identify a_n and b_n.

[0.20 pt]

(c) Under Jeffreys' prior

$$\pi_J(\widetilde{\sigma}) \propto \frac{1}{\widetilde{\sigma}},$$

derive the posterior distribution of $\widetilde{\sigma}$.

Explain how this result can be viewed as a limiting case of the Gamma conjugate analysis.

[0.20 pt]

(d) Suppose

$$n = 10, \qquad \mu = 5, \qquad SS_\mu = 18.$$

Find the posterior distribution under

$$\widetilde{\sigma} \sim \text{Gamma}(2, 1)$$

and under Jeffreys' prior.

Find the posterior mean of $\widetilde{\sigma}$ in both cases.

[0.20 pt]

(e) If

$$\widetilde{\sigma} \mid \boldsymbol{x} \sim \text{Gamma}(a_n, b_n),$$

the posterior predictive distribution for a future observation is a Student t distribution.

State its degrees of freedom, location, and scale in terms of a_n and b_n.

Use the posterior from the conjugate prior in part (d) to identify the predictive distribution.

[0.25 pt]

Question 4: Normal Model with Unknown Mean and Variance [1 point]

Suppose

$$X_1, \ldots, X_n \mid \mu, \sigma^2 \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2),$$

and use the conjugate prior

$$\mu \mid \sigma^2 \sim N\left(\mu_0, \frac{\sigma^2}{p_0}\right),$$

$$\sigma^2 \sim \text{IG}\left(\frac{\nu_0}{2}, \frac{SS_0}{2}\right).$$

Define

$$SS = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

(a) Show that the posterior remains in the normal-inverse-gamma family, with updated quantities

$$p_n = p_0 + n,$$

$$\mu_n = \frac{p_0\mu_0 + n\bar{x}}{p_0 + n},$$

$$\nu_n = \nu_0 + n,$$

and

$$SS_n = SS_0 + SS + \frac{p_0 n}{p_0 + n}(\bar{x} - \mu_0)^2.$$

[0.25 pt]

(b) Explain the interpretation of

$$\mu_n = \frac{p_0}{p_0 + n}\mu_0 + \frac{n}{p_0 + n}\bar{x}.$$

What role does p_0 play?

[0.15 pt]

(c) Explain the three components of

$$SS_n = SS_0 + SS + \frac{p_0 n}{p_0 + n}(\bar{x} - \mu_0)^2.$$

In particular, explain why the final term becomes large when the prior mean and sample mean strongly disagree.

[0.20 pt]

(d) Suppose

$$p_0 = 4, \qquad \mu_0 = 10, \qquad \nu_0 = 6, \qquad SS_0 = 30,$$

and the observed data have

$$n = 8, \qquad \bar{x} = 12, \qquad SS = 42.$$

Calculate

$$p_n, \qquad \mu_n, \qquad \nu_n, \qquad SS_n.$$

Hence state the posterior distributions

$$\mu \mid \sigma^2, \boldsymbol{x}$$

and

$$\sigma^2 \mid \boldsymbol{x}.$$

[0.20 pt]

(e) The marginal posterior of μ is

$$\mu \mid \boldsymbol{x} \sim t_{\nu_n}\left(\mu_n, \sqrt{\frac{SS_n}{p_n \nu_n}}\right).$$

Use the values from part (d) to state this marginal posterior.

Explain why the marginal posterior for μ is a Student t rather than a normal distribution when σ^2 is unknown.

[0.20 pt]

Question 5: Posterior Simulation and Prior Sensitivity [1 point]

Consider the data

```
x <- c(8.4, 10.1, 9.7, 11.2, 10.5,
       9.1, 10.8, 11.0, 9.6, 10.3)
```

and the normal model

$$X_i \mid \mu, \sigma^2 \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2).$$

Use the conjugate prior

$$\mu \mid \sigma^2 \sim N\left(\mu_0, \frac{\sigma^2}{p_0}\right), \qquad \sigma^2 \sim \text{IG}\left(\frac{\nu_0}{2}, \frac{SS_0}{2}\right),$$

with

$$\mu_0 = 9, \qquad p_0 = 3, \qquad \nu_0 = 4, \qquad SS_0 = 8.$$

(a) Calculate

$$n, \qquad \bar{x}, \qquad SS = \sum_{i=1}^{n}(x_i - \bar{x})^2,$$

and then calculate

$$p_n, \qquad \mu_n, \qquad \nu_n, \qquad SS_n.$$

[0.15 pt]

(b) Generate 50,000 posterior draws using the following hierarchical representation:

$$\sigma^2 \mid \boldsymbol{x} \sim \text{IG}\left(\frac{\nu_n}{2}, \frac{SS_n}{2}\right),$$

followed by

$$\mu \mid \sigma^2, \boldsymbol{x} \sim N\left(\mu_n, \frac{\sigma^2}{p_n}\right).$$

Write the R code used for the simulation.

[0.20 pt]

(c) Using the posterior draws, estimate

$$\text{E}[\mu \mid \boldsymbol{x}],$$

$$\text{P}(\mu > 10 \mid \boldsymbol{x}),$$

and a 95% equal-tailed posterior credible interval for μ.

[0.20 pt]

(d) Estimate the posterior mean of σ^2 and a 95% credible interval for σ^2.

Plot the posterior densities of μ and σ^2.

[0.20 pt]

(e) Repeat the analysis using the weaker prior

$$\mu_0 = 0, \qquad p_0 = 0.01, \qquad \nu_0 = 0.01, \qquad SS_0 = 0.01.$$

Compare the posterior means and credible intervals for μ under the two prior specifications.

Briefly explain whether the observed data appear strong enough to reduce the effect of the prior.

[0.25 pt]

End of Homework